SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer **ALL** questions in this section. Marks are indicated at the end of each question. Together they are worth 80% of the total marks for this examination.

Question 1 (28 marks – approximately 51 minutes)

Vienna Manufacturing Limited has two production departments (PD1 and PD2) and two service departments (SD1 and SD2). The two service departments provide supportive services to the two production departments. The budgeted information for the coming year is given below.

Departments:	PD1	PD2	SD1	SD2	Total
Allocated and apportioned overheads	HK$500,000	HK$250,000	HK$120,000	HK$90,000	HK$960,000
Machine hours	400,000	20,000	-	-	420,000
Direct labour hours	10,000	150,000	-	-	160,000
Labour rate per hour	HK$60	HK$55	-	-	-
SD1 charged to	30%	50%	-	20%	100%
SD2 charged to	65%	35%	-	-	100%

Required:

(a) **Prepare a statement showing the apportionment of the overhead costs of the TWO service departments to each production department and the ultimate budgeted overhead costs of each production department.**

(4 marks)

(b) **Determine and explain the appropriate base for absorbing the budgeted overheads of each production department.**

(4 marks)

(c) **Based on your answer to Question 1(b) above, calculate the appropriate budgeted overhead absorption rate (rounded to the nearest cent) for each production department for the coming year.**

(4 marks)

(d) **The company is asked to quote a price for an order. The order requires direct material of HK$200,000, 1,000 direct labour hours in PD1, 10,000 direct labour hours in PD2, and 20,000 machine hours in PD1. It is the company's policy to maintain a uniform gross profit margin of 20% for all sales. Calculate the selling price quoted for the order.**

(6 marks)

(e) **At the end of the coming year, 380,000 machine hours were actually worked in PD1 and 180,000 direct labour hours were worked in PD2. The actual overheads for PD1 and PD2 were HK$560,000 and HK$475,000, respectively. Calculate the amount of under- or over-absorbed overhead for each production department.**

(6 marks)

(f) **Explain, with TWO reasons, why many companies use pre-determined overhead absorption rate, but not actual overhead absorption rate, for product costing.**

(4 marks)

Question 2 (24 marks – approximately 43 minutes)

Oceania Manufacturing Limited ("OML") produces a single product "Gaea" using two types of raw materials, "Agul" and "Belki". OML is preparing its budgets for the coming year. The estimated figures for the coming year include the following:

(1) Production and sales:

	Gaea
Budgeted sales in units	8,200
Opening inventory in units	1,000
Budgeted closing inventory in units	1,800

Normal losses occur at the end of the production process and 10% of the outputs have to be scrapped with zero disposal value. There is no opening or closing work-in-progress.

(2) Direct materials:

	Agul	Belki
Raw materials per unit of production of Gaea	20 kg	10 kg
Opening inventory of raw materials	10,000 kg	12,000 kg
Budgeted closing inventory of raw materials	13,000 kg	11,000 kg
Budgeted purchase price per kg	HK$15	HK$10

The budgeted purchase prices per kg of Agul and Belki for the coming year are the same as those of their actual purchase prices per kg in the current year.

(3) Direct labour:

Standard direct labour time required for producing one unit of Gaea is 1.5 hours. Labour is paid at a rate of HK$60 per hour.

(4) Factory overheads:

The budgeted factory overheads for the coming year are HK$600,000, which are to be absorbed to the production cost on a direct labour hour basis.

Required:

For OML, prepare the following budgets (a) to (e) for the coming year:

(a) Production budget (in units) for Gaea;

(3 marks)

(b) Direct material purchase budget for each of the TWO materials;

(6 marks)

(c) Direct labour budget;

(2 marks)

(d) Pre-determined factory overhead absorption rate; and

(1 mark)

(e) Budgeted factory cost of goods produced showing all cost elements and the unit production costs (rounded to the nearest dollar) for Gaea.

(6 marks)

(f) A proposal was submitted to the Budget Committee of OML to ask for funding to develop a strategic management accounting system. The sales director has asked you, the management accountant, to explain to him the difference between management accounting and strategic management accounting.

(6 marks)

<u>Question 3</u> (14 marks – approximately 25 minutes)

Quinn Industrial Limited ("QIL") has estimated the following annual costs in producing 10,000 units of a component for its production of wireless electric vacuum cleaners for the coming year.

Direct materials	HK$320,000
Direct labour	HK$650,000
Variable production overheads	HK$200,000
Fixed costs	HK$420,000

An outside supplier has offered to supply QIL the required volume of the component at the unit price of HK$140. If the component is purchased from the outside supplier, the facilities where the component is currently made will be laid idle. HK$300,000 of the fixed costs would be incurred no matter if components are made in-house or supplied from outside the company.

<u>Required:</u>

(a) Justify whether QIL should accept the outside supplier's offer of the anticipated production units of the component for the coming year based on financial consideration only.

(6 marks)

(b) If the facilities where the component is currently made can be leased out for HK$150,000 for the coming year, analyse, by showing your working, your new recommendation if different from that in Question 3(a).

(4 marks)

(c) Explain TWO non-financial factors QIL should consider before making this "make or buy decision" for the component.

(4 marks)

<u>Question 4</u> (14 marks – approximately 25 minutes)

Kyle Automation Investment Limited ("KAIL") engages in the distribution and system integration of industrial automation products and information technology products. It has two autonomous divisions: Zhuhai Division and Chongqing Division. The company's reported financial results for last year are as follows:

	Zhuhai Division	Chongqing Division
	HK$	HK$
Total assets	3,600,000	2,200,000
Current liabilities	2,100,000	900,000
Operating income	250,000	180,000

<u>Required:</u>

(a) Calculate the return on investment in percentage (rounded to two decimal places) and the residual income for each division using operating income and investment equal to total assets minus current liabilities. Assume that the required rate of return on investment is 15%.

(8 marks)

(b) Based on your answer to Question 4(a) above, analyse the performance of each division from the point of view of management of KAIL.

(6 marks)

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